Malkkeum App — Product Spec

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Build a Korean mobile app where customers book one-time home cleaning services, pay in the app, and talk to a consultation team by chat for jobs that need a quote.

Overview

Malkkeum (“말끔”, “spotless”) is a consumer app for booking professional cleaning at home in South Korea. Customers are individuals in apartments, studios and houses; the UI language is **Korean**.

- **What customers do:** pick one of 8 cleaning services, choose an address, date and time, then either pay immediately or request a quote through chat.
- **What the business does:** receives bookings, answers quote requests in chat, confirms jobs, assigns a cleaning team, marks jobs done.
- **What you are asked to build:** the customer mobile app, its backend, and a simple admin web panel for staff (the admin panel does not exist in the prototype).
- **All services are one-time bookings.** There are no subscriptions, recurring bookings, memberships or an in-app shop.

The prototype

The clickable prototype is the visual and behavioral reference: match its screens, order of steps and rules, and treat its colors, icons and images as placeholders.

- **How to use it:** open the link on desktop (it shows a phone frame) or on a phone (full screen). Tap through Home → a service → booking, then check My Bookings and the Consultation tab. Bookings, chats, coupons and addresses you create are kept in your browser; “More → 버전 정보 (version)” resets them.
- **Good test paths:** (1) Studio cleaning end to end with a registered card; (2) Air-conditioner cleaning with the 2-in-1 option; (3) Move-in cleaning request, then wait 4 seconds for the chat reply and send “예약 확정할게요”; (4) edit the address to 15 pyeong and try Studio cleaning.
- **Source:** the prototype is one HTML file with plain JavaScript; ask for the file if you want to read field names and pricing logic.

What is fake in the prototype and must be built for real

Prototype behavior	Real requirement
No login; a sample user “홍길동”	Sign-up and login (see Decisions)
Data saved in the browser only	Backend database and API
Payment methods “register” instantly; payment always succeeds	Payment gateway with billing keys, failures, refunds
Chat replies are canned keyword answers	Real staff replying from the admin panel, real-time delivery, push
Status changes via a “[데모]” link	Status changes by staff in the admin panel
Address typed by hand	Address search API
Full time slots are pseudo-random	Real slot capacity
Photo button only shows a message	Photo upload in chat
Emoji and colored blocks as images	Real illustrations and photos

Scope

v1 covers booking, payment, consultation chat and an admin panel; anything not listed under “In” is out unless agreed in writing.

	Item
In	Customer app: Home, My Bookings, Consultation (chat), More tabs and every screen in the Screens section
In	Account: sign-up / login, profile, saved addresses
In	Booking for the 8 services (fixed price, unit-based, quote-based)
In	Card / easy-pay payment through a Korean payment gateway, coupons
In	Real-time consultation chat between customer and staff
In	Push notifications (new chat message, booking status change)

	Item
In	Admin web panel: bookings, statuses, quotes, chat inbox, coupons, notices, FAQ
Out	Recurring / subscription cleaning, membership program, discount for regular customers
Out	In-app shop, products, cart, mileage
Out	Cleaner-side app (team assignment is done by staff in the admin panel)
Out	Points earning rules (the screen exists, balance stays 0 in v1)
Out	Languages other than Korean

Services and pricing

There are 8 services in three booking types; prices below are the prototype's sample prices in KRW (VAT included) and must be editable in the admin panel.

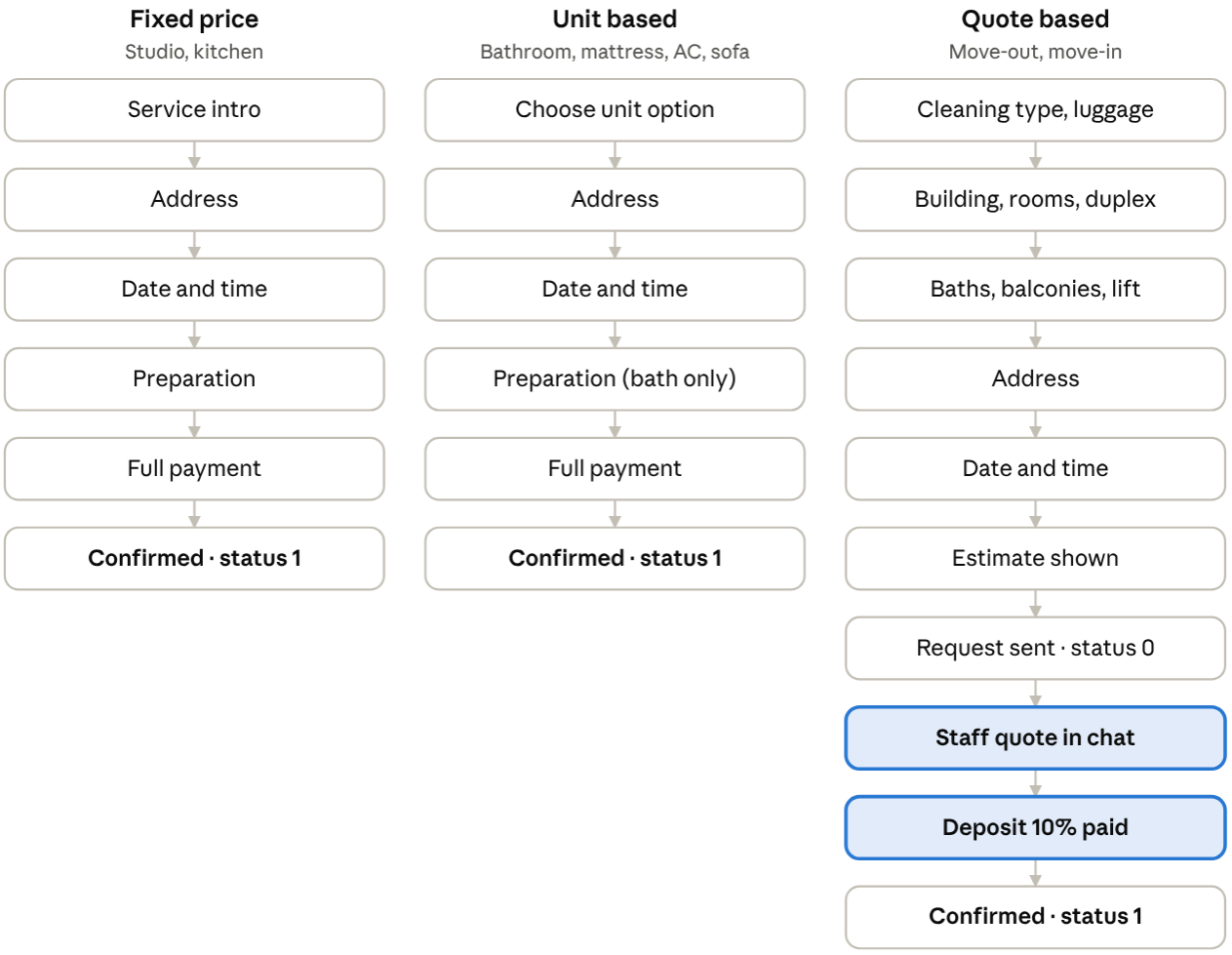
Service (Korean)	Booking type	Customer chooses	Duration	Price (KRW)
Studio cleaning (원룸 청소)	Fixed	— (studios ⩽ 10 pyeong only)	2 h	45,000
Kitchen cleaning (주방 청소)	Fixed	—	2 h	45,000
Bathroom cleaning (화장실 청소)	Unit	Number of bathrooms: 1 / 2 / 3 / 4	2 / 2 / 3 / 4 h	39,000 / 52,000 / 69,000 / 84,000
Mattress cleaning (매트리스 청소)	Unit	Number of mattresses: 1 / 2 / 3 / 4	1 h – 2 h 30 m	59,000 / 99,000 / 139,000 / 175,000
Air-conditioner cleaning (에어컨 청소)	Unit	Wall unit / Stand unit / 2-in-1 / Ceiling unit	1 h 30 m – 3 h	79,000 / 119,000 / 179,000 / 139,000
Sofa cleaning (소파 청소)	Unit	1–2 seat / 3 seat / 4+ seat / L-shape or couch	1 h – 2 h 30 m	59,000 / 79,000 / 99,000 / 119,000

Service (Korean)	Booking type	Customer chooses	Duration	Price (KRW)
Move-out cleaning (이사 청소)	Quote	Cleaning type, luggage state, building, rooms, duplex, bathrooms, balconies, elevator	Set by staff	Estimate, then final quote in chat
Move-in cleaning (입주 청소)	Quote	Same as move-out	Set by staff	Estimate, then final quote in chat

Quote estimate formula (prototype, to be confirmed by the business): base by room layout (studio 180,000 · 1.5-room 230,000 · 2 rooms 300,000 · 3 rooms 380,000 · 4+ rooms 460,000) + 30,000 per bathroom after the first + 15,000 per balcony/utility room + 60,000 if duplex + 40,000 if no elevator and floor ≥ 3, then × (1 + type surcharge + luggage surcharge), then × 1.1 VAT, rounded to 100 won. Type surcharges: after interior work +20%, lived-in deep clean +10%, same-day move +10%, some luggage left +5%. Luggage surcharges: partly unpacked +5%, fully lived-in +15%, unknown +10%.

Booking flows

Fixed and unit jobs are paid upfront; quote jobs are priced in chat first



booking flows · 3 types

Every flow starts from a Home tile and ends in My Bookings. Preparation appears for studio, kitchen and bathroom only. Quote bookings stay in status 0 until staff send a quote in chat and the customer pays the 10% deposit.

Screens

The app has 4 bottom tabs and about 25 screens; the prototype shows each one, so use it as the visual reference and this table as the checklist.

Area	Screen (prototype title)	Purpose	Key elements and actions
Tab 1 · Home	Home (홈)	Entry point	Logo, “all / pick” toggle, bell (→ notices), 4×2 grid of the 8 services, recommended service, auto-rotating benefit banners (4, with counter and list sheet), news list
Tab 2 · My Bookings	Calendar (내 예약)	See bookings by date	Month calendar with prev/next and “today”, dot on dates with bookings, list for the selected date grouped by address, status badge, tap → Booking detail
Tab 3 · Consultation	Chat list (상담)	All chat threads	Thread rows (service icon, title, last message, time, unread badge), “new consultation” button → topic sheet, empty state
Tab 3 · Consultation	Chat room	Talk to staff	Message bubbles, system lines, booking request card, typing indicator, quick replies, text input, photo button, “view booking” link
Tab 4 · More	More (더보기)	Settings hub	Menu groups: service scope, payment methods, coupons, points, account, addresses, notices, tips, FAQ, consultation, partnership inquiry, version, privacy policy, terms, open-source licenses
Booking	Service intro (fixed)	Studio / kitchen info	Description, hero image, duration and price box, included services (studio)
Booking	Unit selection	Bathroom / mattress / AC / sofa	Description, horizontal option cards (label, duration, price), notes, price bar

Area	Screen (prototype title)	Purpose	Key elements and actions
Booking	Address select	Pick address	Saved addresses (radio, edit), add-address sheet (name, address, detail, size in pyeong, home/office)
Booking	Date and time	Pick visit slot	Duration (read-only), visit date (calendar sheet), visit time (slot sheet, some slots full), price bar with breakdown sheet, “nearby customers booking” toast
Booking	Preparation	What to prepare	Tools the customer must have ready (vacuum, trash bags, mop, cloth, gloves, scrubber, detergent), link to chat
Booking	Payment	Discounts and pay	Coupon (select sheet), points, payment method (register sheet), price breakdown, “pay X won” button disabled until a method is set
Booking	Booking complete	Confirmation	Check animation, summary, “home” and “view booking”
Quote booking	Cleaning type	Move-out / move-in step 1	3 type options, luggage state radios, before/after review carousel
Quote booking	Basic info	Step 2	Building type, room layout, duplex yes/no (required marker)
Quote booking	Space info	Step 3	Bathroom count 0–5, balcony count 0–5, elevator yes/no, “floor 3 or higher” toggle
Quote booking	Date and time	Step 5 (after address)	Visit date, morning / afternoon slot, cross-sell coupon download

Area	Screen (prototype title)	Purpose	Key elements and actions
Quote booking	Quote method	Step 6	Estimate card with included guarantees
Quote booking	Confirm	Step 7	Summary of every answer with “change” links back to the step, deposit amount, “request booking”
Booking	Booking detail	One booking	Date, time, status message, 4-step progress bar, payment info, service info, address, chat button, cancel (with confirm sheet); quote bookings show a “request received” sheet once
More	Service scope	What is / isn’t included	Service chips, floor plan with highlighted rooms (or service image), area tabs, included checklist, collapsible “not included” list
More	Payment methods	Register methods	Home / office tabs, method rows with “registered” badge
More	Coupons	Owned coupons	Coupon cards (amount, name, service, expiry, used state), empty state
More	Points	Balance	Balance, empty history
More	Account	Profile	Name, phone, email, login method
More	Addresses	Manage addresses	Home / office tabs, list, add / edit / delete sheet, empty state
More	Notices, Tips, FAQ, legal pages	Content	Lists, accordion FAQ, static pages (content managed in admin)

The prototype's "[데모] 다음 단계로" (demo: next step) link on Booking detail and the "version → reset demo data" action are testing aids; do not build them.

Business rules

These rules are enforced on the server, not only in the app.

Booking status — every booking moves through 4 states; staff change them in the admin panel and the customer sees the progress bar.

Status	Korean	Fixed / unit bookings	Quote bookings
0	상담요청 Consultation requested	Not used	Start state after "request booking"
1	예약확정 Confirmed	Start state after successful payment	After quote agreed in chat and deposit paid
2	매칭완료 Team assigned	Staff assigns a team	Staff assigns a team
3	서비스완료 Completed	After the visit	After the visit

Rules

1. Studio cleaning is allowed only when the chosen address is ⩽ 10 pyeong (about 33 m²). Larger addresses show a message with a "consult" button that opens a chat.
2. Visit date: from tomorrow up to 60 days ahead. Time slots for fixed/unit services: 08–09, 09–10, 13–14, 14–15; a slot can be full (capacity per slot is TBD). Quote bookings use morning (08–09) or afternoon (13–14) arrival.
3. Fixed and unit bookings are paid in full at booking. Payment button stays disabled until a payment method is registered.
4. Quote bookings: no payment at request. The final price is agreed in chat; a deposit of 10% of the final price confirms the booking; the balance is paid after the job (method TBD).
5. Coupons: one coupon per booking; a coupon may be limited to one service; amount is capped at the price; each coupon expires 14 days after issue and is single-use. Sources in v1: welcome pack (10,000 studio + 5,000 any service, once per account) and a cross-sell coupon offered during move-in booking (air-conditioner, 7,000).
6. Cancellation: free while status is 0 or until 2 days before the visit; later cancellation fees are TBD. Cancelling posts a system line in the related chat.
7. Every service needs a saved address; size (pyeong) is required on every address.
8. Prices include VAT and are shown in KRW, rounded to 100 won.

Consultation chat

The chat is how quotes are agreed, so in the real app staff answer it from the admin panel; the prototype's replies are canned keyword responses and must be replaced.

When a thread is created

- Customer submits a move-out or move-in request → a thread linked to that booking opens automatically with a system line ("consultation received") and a booking request card (date, type, layout, estimate, "view booking"). The customer is notified when staff reply.
- Customer taps "consult" on Booking detail → opens the booking's thread (creates one if missing).
- Customer taps "new consultation" and picks a topic (one per service + "other") → a general thread, not linked to a booking.
- Studio booking blocked by the 10-pyeong rule, or the "ask about tools" link on Preparation → a general thread with that topic.

Behavior to build

- Real-time delivery (WebSocket or a chat SDK), message history, read state, unread badge on the tab and per thread, push notification when the app is in the background.
- Message types: text, system line, booking card, photo (customer can send photos of the home; the prototype only shows a button).
- Quick-reply chips above the input: on an open quote thread "confirm booking", "ask about price", "change date", "parking"; otherwise generic ones. They send their text as a normal message.
- Staff can send a **quote message** from the admin panel (final price + deposit). When the customer accepts and pays the deposit, the booking moves to status 1 and a system line appears. In the prototype this is simulated by typing "예약 확정할게요".
- Header shows "Malkkeum consultation team" and the expected response time; business hours (prototype: weekdays 09:00–18:00) and an off-hours auto-message are TBD.

Data model

Seven core entities; the prototype stores the same shapes in the browser, so its source code is a working reference for field names.

Entity	Key fields	Notes
User	id, name, phone, email, login provider, created_at	Phone verified at sign-up (method TBD)
Address	id, user_id, kind (home / office), label, road address, detail, size_pyeong	size_pyeong required; drives the studio rule
Service	id, name_ko, type (fixed / unit / quote), description, duration, price, units[] (label, duration, price), notes[]	Editable in admin; units only for type = unit
Booking	id, user_id, service_id, address snapshot, visit_date, time_slot, status (0–3), unit_index, answers (quote steps), estimate, final_price, deposit, paid_amount, coupon_id, payment_id, chat_id, created_at	Store an address snapshot so later address edits do not change past bookings
Coupon	id, user_id, name, amount, service_id (nullable), issued_at, expires_at, used_booking_id	One per booking
PaymentMethod / Payment	billing key from the gateway, method name; payment: id, booking_id, amount, type (full / deposit / balance), status, gateway transaction id	Never store card numbers; the gateway holds them
ChatThread / Message	thread: id, user_id, booking_id (nullable), topic, unread counts; message: id, thread_id, sender (customer / staff / system), type (text / card / photo / quote), body, created_at, read_at	Quote messages carry price and deposit

Content entities managed in admin: Notice, Tip article, FAQ, Banner (title, subtitle, image, link target, order, active dates), Service scope checklists.

Backend, admin panel and integrations

Nothing behind the prototype is real; all of the following must be built or integrated.

Admin web panel (staff)

- Bookings: list and calendar views, filter by service / status / date, booking detail, change status, assign team (free-text team name in v1), cancel and refund.
- Quotes: see the customer's answers and estimate, send a final quote in the chat, mark deposit received.
- Chat inbox: all threads, unread first, reply, send photos and quote messages.
- Catalogue: edit services, unit options, prices, durations, time-slot capacity.
- Marketing: coupons (create, issue to a user or campaign), banners, notices, tips, FAQ, service scope checklists.
- Staff accounts with at least two roles (admin, consultant).

Integrations (Korea-specific)

Need	Typical options in Korea	Notes
Payment gateway	Toss Payments, PortOne (formerly I'mport), KG Inicis	Card + easy pay (KakaoPay, Naver Pay, Toss Pay); billing key for saved methods; partial refunds
Address search	Kakao (Daum) postcode, Ministry of the Interior road-name address API	Returns road address; customer types detail and size
Login	Kakao, Naver, Apple sign-in, phone number verification	Apple sign-in is mandatory on iOS if other social logins exist
Chat	Build on WebSocket, or a chat SDK such as Sendbird / Channel Talk	Needs staff-side inbox (admin panel)
Notifications	FCM / APNs push; Kakao AlimTalk for booking confirmations	AlimTalk needs a business channel and approved templates
Hosting	Your choice	Personal data must follow Korea's Personal Information Protection Act (PIPA)

Choices marked as options are not decided yet; see the next section.

Decisions needed

The client will settle these with you before the estimate is final; please give a recommendation and cost for each open item.

#	Decision	Options	Status
1	Platforms	iOS + Android (Flutter or React Native) / iOS only / mobile web	TBD
2	Login methods	Kakao, Naver, Apple, phone verification	TBD
3	Payment gateway	Toss Payments / PortOne / KG Inicis	TBD
4	How the quote balance is paid	In-app after the job / on site	TBD
5	Chat technology	Custom WebSocket / chat SDK	TBD
6	Time-slot capacity per day and area	Fixed number per slot / managed in admin	TBD
7	Cancellation fees after the free window	Amounts and timing	TBD
8	Service area	Nationwide / selected regions (block other addresses)	TBD
9	Final prices and quote formula	Prototype values are samples	TBD
10	Brand assets	Logo, icons, illustrations, photos (prototype uses placeholders)	TBD
11	App UI language	Korean only	Decided

Milestones and acceptance criteria

Three milestones, each accepted when every criterion below passes on a real iOS and Android device against a staging backend; dates are set with you after the estimate.

1. **M1 — Accounts and booking with payment**
   - A new user signs up, adds an address via address search, and books each fixed and unit service end to end.
   - The studio rule blocks addresses over 10 pyeong and opens a chat.
   - Full payment succeeds with the gateway's test mode; the booking appears in My Bookings on the right date with status “Confirmed”.
   - Welcome coupons are issued once and reduce the price; a used coupon cannot be reused.
2. **M2 — Quote bookings and consultation chat**

- A move-out and a move-in request create a booking in status 0 and a linked chat thread with the request card.
- Staff reply from the admin inbox; the customer receives the message in real time and as a push when the app is closed.
- Staff send a quote; the customer pays the 10% deposit and the booking moves to “Confirmed”.
- General threads can be started from the topic sheet; unread badges are correct.

3. **M3 — Admin, content and release**
   - Staff change statuses 0–3, assign a team, cancel and refund; the customer sees each change and gets a notification.
   - Services, prices, banners, notices, tips, FAQ and coupons are editable without a code change.
   - The app passes App Store and Google Play review; crash-free sessions ⩾ 99% during a one-week beta.

Glossary

Korean	Meaning
평 (pyeong)	Korean floor-area unit, 1 pyeong ≈ 3.3 m². Addresses store size in pyeong.
원룸 (one-room)	Studio apartment: one room plus a bathroom, usually ⩽ 10 pyeong.
1.5룸	Studio with a separate small room or living area.
이사 청소	Move-out cleaning: done after moving out, usually with the home empty.
입주 청소	Move-in cleaning: done before moving into a new home, often after interior work.
복층	Duplex (two-level) home.
베란다 / 다용도실	Enclosed balcony / utility room, common in Korean apartments.
오피스텔	Officetel: a studio-type unit in a mixed residential/office building.
빌라	Villa: a low-rise apartment building (not a detached villa).

Korean	Meaning
예약금	Deposit paid to confirm a quote booking (10%).
상담	Consultation (the chat with staff).
알림톡 (AlimTalk)	KakaoTalk business notification message.